

The Multinomial Distribution

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Introduction

The multinomial distribution generalises the binomial from two categories to k : the counts of outcomes falling into each of several categories across n independent trials. It underlies the analysis of categorical variables with more than two levels – disease stages, treatment responses, species counts – and sits at the heart of chi-squared goodness-of-fit tests.

Prerequisites

Binomial distribution.

Theory

Let $(X_1, \dots, X_k)$ count outcomes across k categories in n independent trials, where the probability of outcome j on any trial is p_j , and $\sum p_j = 1$.

PMF:

$$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{j=1}^k p_j^{x_j},$$

for $x_j \geq 0$ and $\sum x_j = n$.

Moments:

- $E[X_j] = np_j$.
- $\text{Var}(X_j) = np_j(1 - p_j)$.
- $\text{Cov}(X_j, X_\ell) = -np_j p_\ell$ (negative because one count gain forces another's loss).

Marginals: $X_j \sim \text{Binomial}(n, p_j)$. Pairwise joint for (X_j, X_ℓ) : a trinomial with $(p_j, p_\ell, 1 - p_j - p_\ell)$.

MLE of $(p_1, \dots, p_k)$ is the vector of observed proportions $\hat{p}_j = x_j/n$.

Assumptions

- Fixed total n trials.
- Independent trials.
- Constant category probabilities across trials.

R Implementation

```
set.seed(2026)

p <- c(0.40, 0.30, 0.20, 0.10)
n <- 200

# Single realisation
```

```

one_draw <- rmultinom(1, size = n, prob = p)
data.frame(category = 1:length(p), count = as.integer(one_draw),
            expected = n * p)

# Simulate many replicates, check moments
many <- rmultinom(1e4, n, p)
rowMeans(many); apply(many, 1, var)

# Expected variance
n * p * (1 - p)

# Chi-squared goodness-of-fit
obs <- c(82, 66, 34, 18)
chisq.test(obs, p = p)

# Multinomial likelihood of observed counts
dmultinom(obs, size = sum(obs), prob = p)

```

Output & Results

	category	count	expected
1	1	77	80
2	2	66	60
3	3	38	40
4	4	19	20

```

rowMeans: 80.07 60.04 39.95 19.94
apply var: 48.2 42.0 32.1 17.8

```

```

Expected var:
[1] 48.00 42.00 32.00 18.00

```

Chi-squared test for given probabilities

```

data: obs
X-squared = 3.2, df = 3, p-value = 0.363

```

The empirical means and variances match the theoretical values. The chi-squared test does not reject the specified category probabilities.

Interpretation

Reporting multinomial fits usually focuses on category proportions with uncertainty. For large samples, the chi-squared goodness-of-fit test is the standard tool for comparing observed multinomial counts to hypothesised probabilities.

Practical Tips

- For small expected counts (<5), use Fisher's exact test or Monte Carlo chi-squared (`chisq.test(..., simulate.p.value = TRUE)`).
- Multinomial categories are mutually exclusive by definition; overlapping classifications require a different framework (multilabel models).
- Conditional on the total, the multinomial is a sufficient statistic for the unordered category counts.

- Bayesian inference for multinomial probabilities uses the Dirichlet prior (conjugate); posterior is Dirichlet with shape parameters updated by observed counts.
- For regression with multinomial outcomes, use multinomial logistic regression (`nnet::multinom`).

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots